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EDUCATION

BSc in Electrical and Electronics Engineering

Pamukkale University, Denizli (2021-Present)

GPA: 3.08

PROJECTS

Fault Detection in Induction Motors Using Deep Learning Models in Embedded Systems (BSc Final Project)

- Designed a distributed embedded architecture with **STM32F411** and **STM32F401** microcontrollers.
- Architected multi-threaded firmware on **FreeRTOS**, utilizing queues and semaphores to orchestrate robust synchronization between Data Acquisition, DSP, and Network subsystems.
- Implemented hybrid **1D-CNN and MLP** structures on MCUs for real-time edge computing, achieving **over 90%** fault detection accuracy.
- Developed a multi-threaded **Desktop HIL Interface** using **Python (PyQt6 & pyqtgraph)** capable of real-time TCP/IP data ingestion and spectrogram visualization.

Teknofest Fighter UAV Competition Finalist (2023)

- Developed real-time **PID control algorithms** for autonomous flight stability and target tracking.
- Integrated vision systems with flight controllers, achieving high-precision pursuit accuracy in a constrained embedded environment.

EXPERIENCES

Internship - EÜAŞ 18 Mart Thermal Power Plant (2025)

- Analyzed industrial control loops, SCADA infrastructures, and sensor networks within the power generation units.
- Assisted in the maintenance and diagnostic procedures of large scale synchronous generators.

Internship - KALE Kaleseramik A.Ş. (2024)

- Examined PLC-based automation systems and field instrumentation on active production lines.

SKILLS

Embedded Systems: FreeRTOS, UART, I2C, SPI, TCP/IP, UDP.

Programming: C, Python, MATLAB.

Frameworks: CMSIS-DSP, TensorFlow, OpenCV, PyQt6, pyqtgraph.

Software: Ansys Maxwell 2D/3D, MATLAB/Simulink, MS Office.

LANGUAGES

English – C1